Shaping the battery's bid curves from the multiplier ranges

A proposal for discussion: what it can fix, what it cannot, and how to test it

Companion to `model_structure.pdf` (§7, the multiplier polyhedron)
and `regimes_design.pdf`

Status: idea and diagnosis only; nothing implemented

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In short. The ranges of valid π_t are information the current bid throws away, so turning them into richer bid curves is worth exploring. But a close look at the loss at *perfect* prices (6.32% of V) shows it is an **inter-temporal** problem: an hourly market cannot make a sale at 04–08h conditional on a purchase at 00–04h, and no price inside the polyhedron changes that. Hourly curves are therefore likely to help against **forecast uncertainty** (wrong prices), not against this coordination loss. The ranges fit most naturally into **SOC-dependent bids**, an intermediate design between today's hourly curves and Baldick's SOC-based clearing. Below: the diagnosis, five ideas, the hypotheses each one makes, and a test plan.

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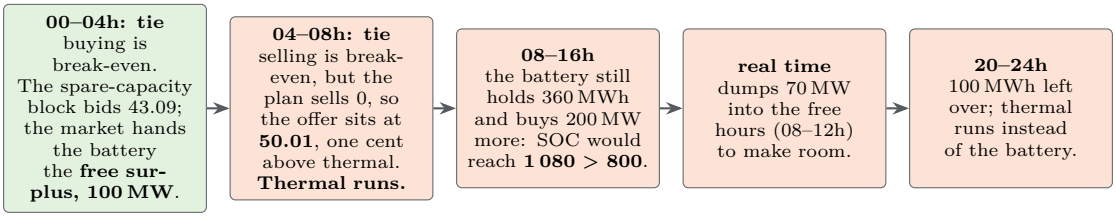
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1 Where the loss at perfect prices comes from

`B_star_p` bids from the *true* prices $\lambda^\star$, with the battery's own plan (its self-schedule) as quantity. It still loses 6.32% of V. Hour by hour (MW; day-ahead = what the market accepted):

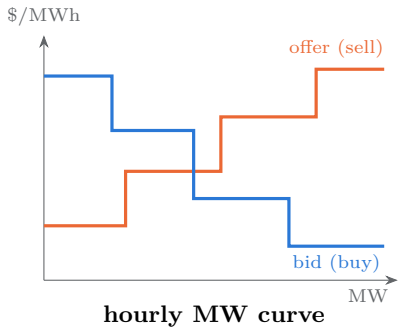
interval	00–04	04–08	08–12	12–16	16–20	20–24
true price $\lambda^\star$ ($/MWh)	43.1	50	0	0	50	50
Baldick (benchmark)	buy 100	**sell 90**	buy 200	buy 22	sell 175	sell 25
battery's plan (self-schedule)	–	–	–	buy 222	sell 200	–
`B_star_p`, day-ahead	buy 100	**sell 0**	–	buy 200	sell 175	–
SOC if delivered (MWh)	360	360	360	**1 080**	380	380
real time	buy 100	–	sell 70	buy 200	sell 175	–
thermal output, real time	0	**100**	0	0	0	**100**
thermal output, Baldick	0	10	0	0	0	75



The cost, item by item (against Baldick): extra thermal generation +23 000 $ (04–08h and 20–24h), re-dispatch premium +2 800 $, less degradation −269 $, value of the 100 MWh left at the end −4 000 $: in total 21 531 $ = 6.32% of V.

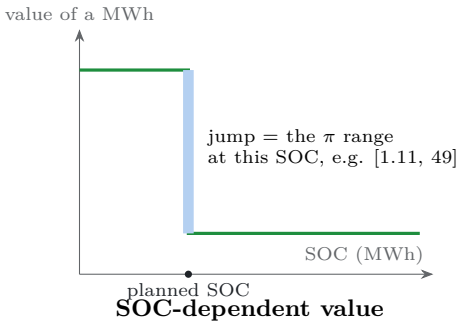
Diagnosis. Buying at 00–04h and selling at 04–08h belong together: each alone is worth nothing to the battery, together they save 90 MW of thermal at 04–08h. The hourly market resolves the two ties *independently*: the first in favour of buying (the battery is the only bidder for the free surplus), the second against selling (thermal at 50 beats the offer at 50.01). Baldick's clearing sees both hours and the SOC at once and resolves them together.

2 What a curve can and cannot express



hourly MW curve
(current design)

One curve per hour: "how much at which price, *in this hour*". It cannot say "sell at 04–08h *only if* you bought at 00–04h".



SOC-dependent value
(the battery's marginal value curve)

The value of a stored MWh as a function of SOC. The π range is the vertical jump at the planned SOC. If the *market* tracks SOC, this curve answers "sell at 04–08h because you are now full".

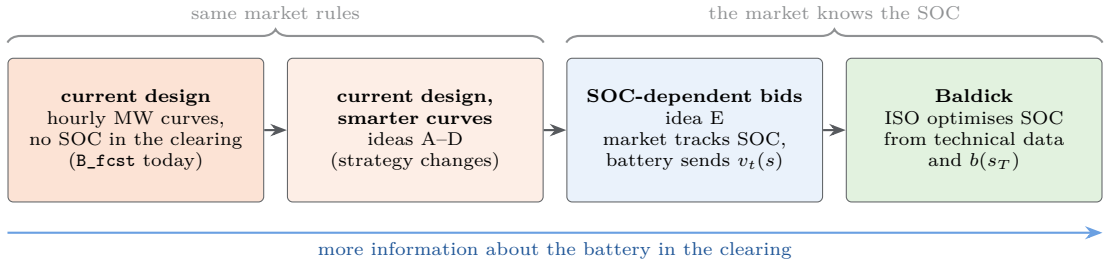
Two consequences.

1. **Choosing inside the range does not fix the coordination loss.** The polyhedron is built around the battery's *plan*. Where the plan does not sell (04–08h), every point of Π has an offer at or above the price, $c_2 \geq \lambda_2^\star = 50$. Any curve that stays inside Π inherits the plan's blind spot. This is Finding 1 of `model_structure.pdf` seen from the other side: all four corners lose the same 6.32%.
2. **The ranges are naturally SOC-dependent information.** A range exists because the planned SOC sits on a limit; it says how the value of energy jumps there. That is exactly what an SOC-dependent bid carries, and what an hourly MW curve has no way to carry.

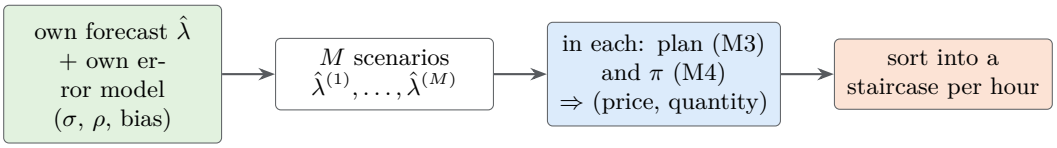
3 Five ideas

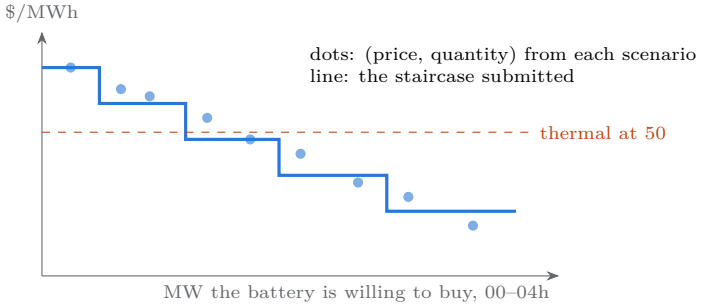
	Idea	How the curve is built	Against ties / coordination	Against forecast error
A	Range-spread MW curve	Price steps across $[\mathrm{lo}_t, \mathrm{hi}_t]$ of π_t (through (E)); the planned quantity at one end, spare capacity stepped toward the other.	unlikely (stays inside Π)	little
B	Value-function MW curve	Dynamic programme against $\hat{\lambda}$: marginal value $v_t(s)$ on an SOC grid; blocks of Δs MWh priced $v_t(s)/\beta + \zeta$ (sell) and $\alpha v_t(s) - \varepsilon$ (buy) around the *planned* SOC. Archived code: `Archive/guide_md_framework/src/dp.py`, `curves.py`.	unlikely (still built around the plan)	little
C	Scenario MW curve	The battery samples M forecasts from *its own* error model around $\hat{\lambda}$, solves its plan and π in each, and sorts the (price, quantity) pairs into a staircase: "buy q MW if the price is at most p_q".	partly	**yes**
D	Stochastic value-function curve	B with the value function averaged over the scenarios of C.	partly	**yes**
E	SOC-dependent bid	The battery submits $v_t(s)$ (its marginal value of energy *as a function of SOC*); the clearing tracks SOC. The π ranges are the jumps of v_t.	**yes**	with D: **yes**

A–D stay within the current design (hourly curves, no SOC in the clearing). They change the *battery's strategy*, so they answer: *how much of the gap to Baldick is due to naive bidding?* **E changes the design**: the market must track SOC. It sits between today's design and Baldick's (where the ISO also optimises the SOC), and it is the direction some operators are taking for storage (see the SOC-awareness and SOC-dependent-bid documents in `Literature/`, e.g. ERCOT NPRR 1186 and the paper on multi-interval co-optimisation with SOC-dependent bids).



4 How idea C would build a curve



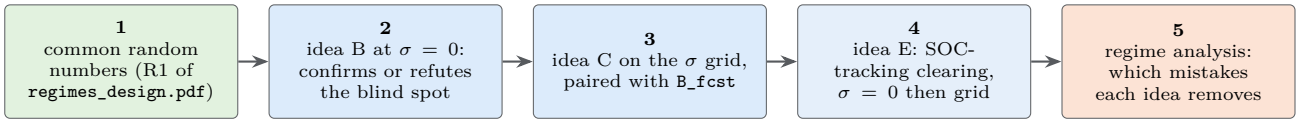


Illustration, not computed. Instead of one price for the whole spare capacity, the battery buys more the cheaper the energy is, in proportion to how many of its scenarios say that purchase pays. Hypothesis: this avoids the expensive "buy from thermal at 00–04h" regime ($\mathcal{R}_5$ in `regimes_design.pdf`), because only a small share of the curve sits above the thermal price.

5 Hypotheses and how to test them

Idea	Hypothesis	Test
A, B	Do *not* remove the 6.32% perfect-price loss (they stay inside the plan's blind spot).	$\sigma = 0$: loss $\approx 6.32\%$
C, D	Reduce the loss under forecast error, mainly by removing expensive regimes; no gain at $\sigma = 0$.	paired comparison with `B_fcst`, same error shapes, $\sigma \in \{1, \ldots, 20\}$
E	Removes most of the perfect-price loss (the market sees the SOC); with D, also part of the forecast loss.	$\sigma = 0$: loss $\to 0$; then the σ grid

How every idea is scored. A new bid family per idea (`B_curve_range`, `B_curve_dp`, `B_curve_scen`, ...) returning more blocks per side; `market.clear_hour` already accepts any number of blocks, so clearing, real time and welfare (`runner.evaluate_blocks`) are reused unchanged. Idea E needs a new clearing variant that carries the SOC across hours (one multi-interval LP instead of six hourly ones). It does not exist yet: the archived framework's `clearing.py` clears each hour independently and only tracks the SOC afterwards; the root `CLAUDE.md` asked for an `soc_enforced` switch, but it was never built. `verify_equivalence.build` already contains the SOC rows (A8)–(A10) that such a variant would reuse.



6 Questions to discuss

1. **Scope.** Is the thesis about *market design* only (then A–D are a robustness check: "the gap survives smarter bidding") or also about *how batteries should bid* in today's design (then C/D are a main result)? Possibly a question for the supervisors.
2. **Idea E as a third design.** Comparing three designs (hourly curves / SOC-dependent bids / Baldick) instead of two would place the thesis in the current operator debate. Is it in scope, and which existing proposal should it follow?
3. **Information assumptions.** Idea C assumes the battery knows its own error statistics; idea E assumes the market accepts an SOC-dependent value curve. Are both acceptable assumptions for the thesis?
4. **Test system.** The coordination loss here is driven by exact ties of a 6-interval case. Should the curve ideas be evaluated directly on a larger, tie-free system (stage 1/3), where the picture may differ?
5. **Order of work.** Proposed: common random numbers first, then the $\sigma = 0$ test of B (one day of work) to confirm the diagnosis before investing in C–E.

Numbers in §1 come from the current code (`bids.bid_star_p`, prelimit rule, `runner.evaluate_blocks`, $\psi = 10$), the same draw as `B_star_p` in the stage-0 notebook. The figures in §2 and §4 are schematic illustrations, not computed results.